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RCPL rPET QR Code Information System

Product Requirements Document

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✓ Approved for Development

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1. Executive Summary

> **Project:** RCPL requires a QR Code-based product information system for their **rPET** product line. A QR code is already purchased from QR Planet. This system provides three independent components: consumer landing page, product information display, and regulatory admin portal. The system must be **fully independent** from existing RLL/RCPM QR systems.

The key deliverable is a **URL** that RCPL's regulatory team maps to the QR Planet-hosted QR code. Consumers scan the QR code on the bottle, enter the batch code (first 1–2 characters printed on the bottle neck), and instantly see manufacturer and FSSAI compliance details on their mobile device.

Metric	Target	Metric	Target
Consumer page load	< 3 seconds on 4G	API response time	< 500ms P95
End-to-end QR flow	< 10 seconds	Admin onboarding	< 30 minutes

2. Problem Statement

Consumer Problem
Consumers have no easy way to verify rPET product authenticity, manufacturer details, or FSSAI compliance. Regulatory requirements mandate this information be accessible via QR code.

Regulatory Team Problem
The existing vendor management system (RLL/RCPM) cannot be modified. A new independent system is needed that the regulatory team can update without IT dependency for each change.

Solution: Three-component web system (Consumer Landing Page → Batch Code Lookup → Admin Data Portal) deployed as an independent URL, mapped to the existing QR Planet QR code.

3. Stakeholders

Stakeholder	Role	Primary Need
Consumers	End users scanning QR code on bottle	Fast, mobile-friendly product info
Nidhi (Regulatory Team)	Data manager — adds/edits rPET records	Simple admin form, no code changes
RCPL IT (Ashcharna/Pankaj)	Technical reference for existing system	Zero changes to RLL/RCPM systems
Ashish Chandorkar	Developer & system owner	Clear requirements, URL to QR Planet
QR Planet	QR code provider	Landing page URL for QR code mapping

4. User Stories

Consumer Stories

ID	As a…	I want to…	So that…
US-01	Consumer	Scan QR and see manufacturer details	I can verify product authenticity
US-02	Consumer	Enter only 1-2 chars from bottle neck	The process is quick on mobile
US-03	Consumer	See FSSAI license, vendor name, address	I have complete compliance information
US-04	Consumer	Page loads fast on mobile data	I don't wait more than 3 seconds

Regulatory Team Stories

ID	As a…	I want to…	So that…
US-05	Regulatory team member	Add new rPET product records	Consumer info stays current
US-06	Regulatory team member	Edit FSSAI license or address	Records are always accurate
US-07	Regulatory team member	Deactivate discontinued products	Outdated products don't show to consumers
US-08	Regulatory team member	Search products by vendor/category	I can find records quickly

5. Functional Requirements

P0 Must Have — Launch Blockers

ID	Requirement
FR-01	Consumer landing page with QR scan instructions (3-5 lines) and batch code input field
FR-02	Batch code entry: 1-10 alphanumeric characters, case-insensitive, auto-uppercase transform
FR-03	Product info display: Product Name, Category, Subcategory, Vendor Name, Vendor Address, Manufacturer, FSSAI License, SKU, Status
FR-04	Admin form to CREATE product records with all required fields and FSSAI validation
FR-05	Admin form to EDIT existing product records (batch_code immutable after creation)
FR-06	Inactive product codes return friendly "information not available" message
FR-07	Non-existent codes return "product not found" with clear guidance
FR-08	System provides a standalone deployable URL for QR Planet mapping

P1 Should Have

ID	Requirement

FR-09	Admin product list with search (vendor name, category, batch code)
FR-10	Paginated admin product list (15 per page)
FR-11	Admin soft-delete (deactivate) with confirmation modal
FR-12	FSSAI validation: exactly 14 digits, numeric only
FR-13	Rate limiting on public endpoint (100 req / 15 min)
FR-14	Health check endpoint at /health

P2 Future Sprints

ID	Requirement
FR-15	Admin portal authentication (login credentials for regulatory team)
FR-16	Audit log of all admin changes (who changed what, when)
FR-17	Bulk import via CSV for initial data migration
FR-18	QR code generation preview in admin portal

6. Non-Functional Requirements

Category	Requirement
Performance	Consumer pages: < 3s on 4G mobile. API lookup: < 500ms P95
Mobile-first	Landing page and product info page optimised for 375px+ (iPhone SE) upward
Security	Helmet.js headers, input sanitisation (express-validator), parameterised queries, rate limiting
Availability	99% uptime; SQLite WAL mode for concurrent reads
Independence	Zero dependencies on existing RLL/RCPM vendor management system
URL Stability	Public URL must not change once mapped to QR Planet
Data Integrity	batch_code is UNIQUE; FSSAI is 14-digit validated; status check constraint

7. System Architecture

```
┌──────────────────────────────────────────────────────┐
│  QR Planet (3rd party)                               │
```

```
│  QR Code ──maps to──▶ https://[your-domain]/              │
└──────────────────────────────────────────────────────────┘
                            │ Consumer scans QR code
                            ▼
┌──────────────────────────────────────────────────────────┐
│  FRONTEND  (React 18 + TypeScript + Tailwind CSS, Vite) │
│  /              ──▶ Landing Page (batch code search)     │
│  /product/:code ──▶ Product Info Page (lookup result)    │
│  /admin/*       ──▶ Admin Portal (regulatory team CRUD) │
└──────────────────────────────────────────────────────────┘
                            │ REST API (JSON)
                            ▼
┌──────────────────────────────────────────────────────────┐
│  BACKEND  (Node.js + Express + Helmet + Rate Limiter)    │
│  GET  /api/products/lookup?code=XX  ◀── Public           │
│  GET/POST/PUT/DELETE /api/admin/*   ◀── Regulatory team  │
│  GET  /health                       ◀── Monitoring       │
└──────────────────────────────────────────────────────────┘
                            │ better-sqlite3
                            ▼
┌──────────────────────────────────────────────────────────┐
│  DATABASE  (SQLite, WAL mode)                            │
│  Table: rpet_products                                    │
│  batch_code (UNIQUE NOCASE), category, vendor_name,      │
│  fssai_license, status, created_at, updated_at, ...      │
└──────────────────────────────────────────────────────────┘
```

8. Data Model

Table: `rpet_products`

Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Internal identifier
batch_code	TEXT	UNIQUE, NOT NULL, COLLATE NOCASE	Printed on bottle neck (e.g. "AA")
category	TEXT	NOT NULL	e.g. "Dairy"
subcategory	TEXT	NOT NULL	e.g. "Full Cream Milk"
vendor_name	TEXT	NOT NULL	Manufacturing unit name
vendor_address	TEXT	NOT NULL	Full address with PIN code
fssai_license	TEXT	NOT NULL	14-digit FSSAI license number
sku	TEXT	nullable	Stock Keeping Unit code

product_name	TEXT	nullable	Full product name
manufacturer	TEXT	nullable	Parent company name
status	TEXT	DEFAULT 'active', CHECK IN ('active','inactive')	Active / Inactive
created_at	TEXT	DEFAULT datetime('now')	ISO 8601 timestamp
updated_at	TEXT	DEFAULT datetime('now')	ISO 8601 timestamp

9. API Specification

Public Endpoint

Method	Path	Auth	Description
GET	/api/products/lookup?code={batchCode}	None	Consumer product lookup by batch code
GET	/health	None	Service health check

Response	Status	Condition
`{ success: true, data: RpetProduct }`	200	Valid active code
`{ success: false, error: "not currently available" }`	400	Code exists but product inactive
`{ success: false, error: "No product found..." }`	404	Code does not exist
`{ success: false, error: "validation message" }`	400	Invalid code format

Admin Endpoints

Method	Path	Description
GET	/api/admin/products?page=1&limit=20	List all products (paginated)
GET	/api/admin/products/search?q=query	Search products
GET	/api/admin/products/:id	Get single product
POST	/api/admin/products	Create product (201)
PUT	/api/admin/products/:id	Update product
DELETE	/api/admin/products/:id	Soft-delete (set inactive)

10. UI/UX Requirements

Page	Key Elements	Design Notes
Landing Page (/)	RCPL logo, 3-step instruction box, batch code input, "Get Product Details" button, footer	Mobile-first, blue gradient bg, white card, large monospace input, auto-uppercase
Product Info (/product/:code)	Back button, batch code display, product name hero, 3 info cards, status badge, loading skeleton	Info cards: Product Details, Manufacturer Details, Compliance. Shimmer skeleton loading.
Admin Portal (/admin/*)	Collapsible sidebar, product table with search+pagination, Create/Edit form, confirmation modal, toasts	Desktop-optimised, responsive on mobile. Blue theme. Inline validation.

11. Success Metrics

Metric	Target	Measurement Method
Consumer page load time	< 3 seconds on 4G	Lighthouse / browser DevTools
API lookup response P95	< 500ms	Server logs / monitoring
QR scan → info display	< 10 seconds	Manual end-to-end test
Admin onboarding time	< 30 minutes	User observation
Admin form error rate	< 2% of submissions	Validation error logs

12. Out of Scope

- Integration with existing RLL/RCPM vendor management system
- Consumer authentication or account creation
- QR code generation (handled by QR Planet)
- Email/SMS notifications
- Multi-language support (Hindi, regional languages)
- Native mobile app
- Price or inventory information

13. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
QR Planet URL mapping delay	Low	High	Provide URL early; test with direct URL first
Regulatory team needs more fields	Medium	Medium	Schema designed with optional fields; easy to add columns
Mobile network failures	Medium	High	Error states with retry; loading skeletons for perceived performance
SQLite concurrency under load	Low	Medium	WAL mode enabled; upgrade to PostgreSQL if traffic grows
Admin portal misuse (no auth)	Medium	Medium	P2: Add login auth in Sprint 2

14. Acceptance Criteria

- ☐ Consumer can scan QR, enter "AA", see full product details within 10 seconds on mobile
- ☐ Inactive product code returns "information not available" (not a 404 error)
- ☐ Non-existent code returns friendly "product not found" with back button
- ☐ Admin can create a new product record with all required fields
- ☐ Admin can edit vendor address and FSSAI license number
- ☐ Admin can deactivate a product (soft delete with confirmation)
- ☐ FSSAI validation rejects anything that is not exactly 14 digits
- ☐ All pages render correctly on iPhone SE (375px) screen width
- ☐ CI/CD pipeline passes: lint → test → build → security audit
- ☐ Landing page URL is provided for QR Planet mapping

15. Delivery Plan

0. **Foundation** — CLAUDE.md, package.json, tsconfig, .gitignore, CI/CD pipeline *(Day 1)*
1. **Backend API** — Express + SQLite + all endpoints (lookup + admin CRUD) *(Day 1)*
2. **Consumer Frontend** — Landing Page + Product Info Page (mobile-first, React + Tailwind) *(Day 1–2)*
3. **Admin Portal** — Product list, create/edit forms, pagination, search *(Day 2)*
4. **Testing** — Unit + integration tests for frontend and backend *(Day 2)*
5. **Security Review + Deployment** — URL delivery to QR Planet team *(Day 3)*

6 **Demo & Feedback** — Regulatory team walkthrough, Nidhi shares final variable list *(Week 2)*